The State of Valuing Urban Ecosystem Services

Andie M. Creel

November 9 2026

Urban Ecosystem Services Workshop - Yale/Gund/NatCap

Lay of the Land

- Urban natural assets (parks, street trees, wetlands, rivers) provide heat control, cleaner air and water, flood mitigation, a place to recreate, and better physical and mental health (Keeler et al., 2019; Baró et al., 2016)

Lay of the Land

- Urban natural assets (parks, street trees, wetlands, rivers) provide heat control, cleaner air and water, flood mitigation, a place to recreate, and better physical and mental health (Keeler et al., 2019; Baró et al., 2016)
- A large and growing literature (Veerkamp et al., 2021; Ma and Yang, 2025,?):
  - Most studied: local temperature regulation (36% of studies) and recreation (23%) (Veerkamp et al., 2021)
  - Large, publicly owned natural assets (*e.g.,* parks) are studied most often
  - Still gaps: Value of benefits (especially private benefits), and smaller or less visible natural assets

Lay of the Land

- Urban natural assets (parks, street trees, wetlands, rivers) provide heat control, cleaner air and water, flood mitigation, a place to recreate, and better physical and mental health (Keeler et al., 2019; Baró et al., 2016)
- A large and growing literature (Veerkamp et al., 2021; Ma and Yang, 2025,?):
  - Most studied: local temperature regulation (36% of studies) and recreation (23%) (Veerkamp et al., 2021)
  - Large, publicly owned natural assets (*e.g.,* parks) are studied most often
  - Still gaps: Value of benefits (especially private benefits), and smaller or less visible natural assets
- Despite size of literature, it is hard to translate this literature into decisions about urban nature

Challenges

- **No shared definition:** Half of >300 reviewed papers never define "ecosystem service" (Tan et al., 2020)
  - Definitions vary across fields, and within them (Boyd and Banzhaf, 2007)

Challenges

- **No shared definition:** Half of >300 reviewed papers never define "ecosystem service" (Tan et al., 2020)
  - Definitions vary across fields, and within them (Boyd and Banzhaf, 2007)
- **No shared method:** At least 47 methods for quantifying urban ecosystem services (Ma and Yang, 2025)

Challenges

- **No shared definition:** Half of >300 reviewed papers never define "ecosystem service" (Tan et al., 2020)
  - Definitions vary across fields, and within them (Boyd and Banzhaf, 2007)
- **No shared method:** At least 47 methods for quantifying urban ecosystem services (Ma and Yang, 2025)
- **Benefits are rarely measured:** 59% of papers quantify ecosystem functions/properties, a minority state the benefit to people, even fewer in dollars (Veerkamp et al., 2021)

Challenges

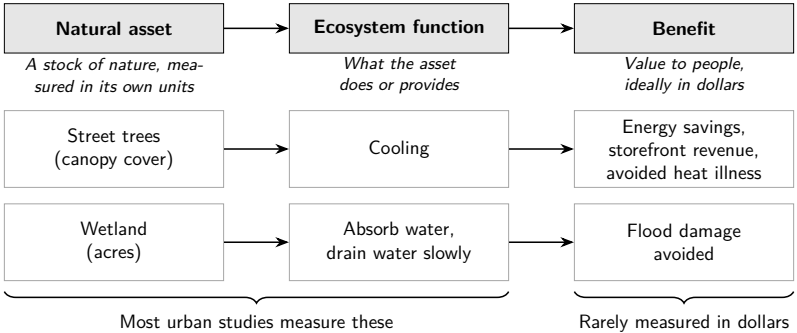
- **No shared definition:** Half of >300 reviewed papers never define "ecosystem service" (Tan et al., 2020)
  - Definitions vary across fields, and within them (Boyd and Banzhaf, 2007)
- **No shared method:** At least 47 methods for quantifying urban ecosystem services (Ma and Yang, 2025)
- **Benefits are rarely measured:** 59% of papers quantify ecosystem functions/properties, a minority state the benefit to people, even fewer in dollars (Veerkamp et al., 2021)
- ***Why dollars matter:***
  (spoiler, it's not to sell ecosystem services to highest bidder)
  - A city needs to know value of ecosystem services for benefit-cost analysis (trees vs. schools, roads, safety)
  - Quantifying private benefit (increased revenue, cost savings) can mobilize private money
  - A non-profit can use valuation studies to tell funders what a dollar spent on nature returns

Consensus is hard to build when you talk past each other

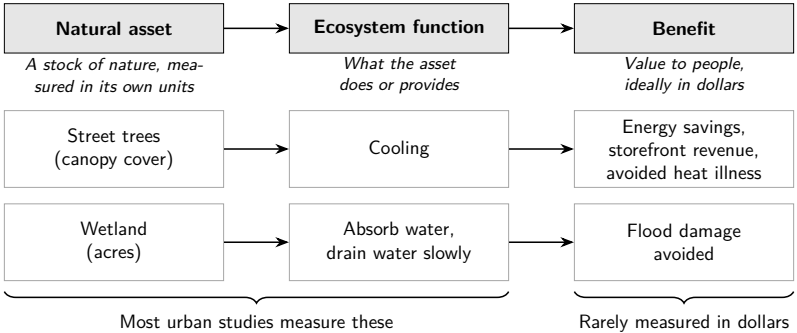
- Each of these three papers are cited 2,700–4,200 times on Google Scholar
- Flood prevention is a service in one paper and a benefit in another

	Haines-Young and Potschin (2010)	Gómez-Baggethun and Barton (2013)	Boyd and Banzhaf (2007)
Wetland	Ecosystem (structure)		Service
Water slowing or soaking into soil	Function and a service	Function	Input to services
Flood control	Justification for why the function counts as a service	Service	Benefit

We Want To Communicate! – Three Key Concepts

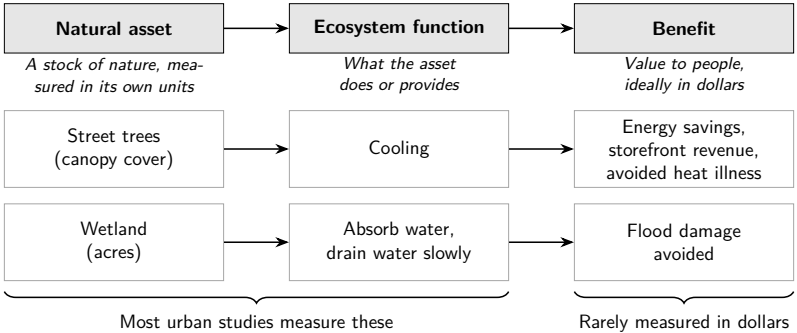


We Want To Communicate! – Three Key Concepts



- Natural capital accounting puts a dollar value on a unit of the asset, but it is technically hard (Fenichel, 2024)
- **Quantifying benefits perfectly should not be the enemy of measuring them at all!**

We Want To Communicate! – Three Key Concepts



- Natural capital accounting puts a dollar value on a unit of the asset, but it is technically hard (Fenichel, 2024)
- **Quantifying benefits perfectly should not be the enemy of measuring them at all!**
- Cultural and spiritual services (*e.g.*, a river's spiritual value) are important, but hard to price (Maher et al., 2020; Boyd and Banzhaf, 2007); not my expertise

Many natural asset's functions are fairly well documented

Table 1 | Urban ecosystem services and nature-based solutions

Urban ecosystem services	Nature-based solutions						
	Street trees	Parks and open space	Engineered stormwater controls[a]	Green roofs	Waterways and wetlands	Coastal habitats[b]	Urban gardens
Urban air quality	X	X					
Carbon sequestration	X	X					
Coastal protection						X	
Urban heat and heat extremes mitigation	X	X		X			
Stormwater and wastewater management	X	X	X	X	X		
Urban water supply	X	X	X				
Riverine flood impact reduction		X	X		X		
Recreation opportunities		X			X	X	
Mental health	X	X					X
Urban agriculture		X					X

We reviewed literature pertaining to ten ecosystem services associated with seven nature-based solutions. Gaps in this matrix identify combinations where there was not sufficient literature to review the impact of a specific solution on a given service and/or the nature-based solution has been determined to be minimally effective with respect to a particular ecosystem service. [a]Such as bioswales, rain gardens, retention ponds. [b]Such as oyster reefs, mangroves, dunes, marshes.

Notes: Table 1 of Keeler et al. (2019) in *Nature Sustainability*. Their nature-based solutions are my natural assets; their urban ecosystem services are my ecosystem functions.

Heat mitigation: the benefit of cooling has been valued in several ways

- **Natural assets:** trees, parks, green roofs, green walls
- **Function:** cooling through shade and evapotranspiration (Mohajerani et al., 2017; Wong et al., 2021)
  - Greenery lowers peak surface temps 2–9 °C (Wong et al., 2021)
  - Phoenix: vegetation cools the surface by nearly 25 °C vs. bare soil (Jenerette et al., 2011)
- **Benefit:** lower energy bills, steadier business revenue on hot days, increased health
- **What we know in dollars:**
  - Phoenix households would pay \$72/month for 1 °F cooler (Klaiber et al., 2017)
  - Trees in major U.S. cities saved \$539M in cooling energy in 2016 (Heris et al., 2021)
  - Green space around a business raises revenue 0.5 pct. points on days $\geq$100 °F (Creel, 2026)
- **What we don't know:** Value of health benefits of cooling, such as reduced heat-related illness and mortality (more later), private incentive to invest in cooling infrastructure

Recreation: valued in dollars, but how much is due to nature is unclear

- **Natural assets:** parks, trails, water bodies, tree-lined sidewalks
- **Function:** a place to recreate and spend leisure time outside
  - Most U.S. outdoor recreation is local: walking, playing with kids or pets (Creel and Fenichel, 2026)
  - Poorly understood function: Parks also build social ties among neighbors (Kaźmierczak, 2013; Peters et al., 2010). Do they increase social cohesion?
- **Benefit:** the value people get from leisure time outside
- **What we know in dollars:**
  - Local outdoor recreation: \$5–\$6 per trip, ≈\$93B/year (Creel and Fenichel, 2026)
  - Lower crime "unlocked" ≈\$7B in property value near parks (Albouy et al., 2020)
  - Germans would pay €20.25/month for more natural green space (Bronnmann et al., 2023)
- **What we don't know:** how much visits are driven by various amenities of a park; the value of social ties
  - NYC: fewer social media posts in parks with more green space (Hamstead et al., 2018)

Stormwater: well studied, but its dollar value leaves out flood damage

- **Natural assets:** trees, parks, wetlands, and other permeable surfaces (rain gardens, swales)
- **Function:** intercept and absorb rainfall, so less runoff (Veerkamp et al., 2021; Heris et al., 2021)
  - Unclear how well each assets reduce flooding (Diep and McPhearson, 2025); trees work best in small storms (Heris et al., 2021)
- **Benefit:** less flood damage, lower wastewater treatment costs, fewer sewage overflows
- **What we know in dollars:** limited work
  - Trees in major U.S. cities intercepted 2.6B m^3 of rain in 2016, worth \$425M (Heris et al., 2021)
  - Avoided treatment costs (\$2.58/m^3), only in cities with combined sewers
- **What we don't know:** the value of avoided flood damage to homes and businesses

Health has been valued mainly through avoided heat deaths

- **Natural assets:** parks, trees, and green space more broadly
- **Function:** Many functions contribute to health. Health is a benefit of cooling, cleaner air, and a place for activity, social contact, and rest from stress (Jennings and Gaither, 2015; Keeler et al., 2019)
- **Benefit:** fewer heat- and pollution-related deaths and hospital visits; better mood/lower stress
  - In less green ZIPs, heat raises odds of elderly cardiovascular death 17% (Gronlund et al., 2015)
  - Losing 100M ash trees $\rightarrow$ 15,080 more cardiovascular deaths (Donovan et al., 2013)
  - Green space use tied to better well-being (Jennings and Gaither, 2015)
  - Heat raises mental health helpline calls 3.7% (Janzen, 2025)
  - Mostly observational and confounded by wealth (Donovan et al., 2013; Ogletree et al., 2023)
- **What we know in dollars:** tree cover in 97 U.S. cities avoids $\approx$296 heat deaths/year, worth \$1.0–\$2.4B/year (McDonald et al., 2020)
- **What we don't know:** How much money insurance companies save from avoided heat-related health damage

Natural assets are unequally distributed

- Fewer natural assets:
  - Historically redlined neighborhoods have less tree canopy (Locke et al., 2020)
  - Less affluent neighborhoods have less protected land(Sims et al., 2022)
- More heat:
  - Historically redlined neighborhoods are hotter (Hoffman et al., 2020; Wilson, 2020; Conzelmann et al., 2022)
  - Black residents are more likely than white to live with little tree canopy and mostly impervious surface (Jesdale et al., 2013)
- Benefit valuation across demographics is sensitive to the method used:
  - Valuing travel time with a replacement wage instead of 1/3 own wage: richest quintile's value per trip goes from 10× the poorest's to 1.5× (Creel and Fenichel, 2026)
  - Tree cover where people shop is nearly uncorrelated with customers' income (Creel, 2026)

Urban animals: the next frontier of ecosystem service measurement?

- Big gap: function of animals in cities
  - Review of 317 papers finds only 2 studies on how animal traits affect an ecosystems' functions (Schwarz et al., 2017)
  - Outside cities: bats and birds control pests → big cost savings for farmers (Frank, 2024; Noack et al., 2025), wolves cut deer-vehicle collisions, saving >$375K per county per year (Raynor et al., 2021)
- Possible urban services:
  - Coyotes and foxes may keep rats in check ⇒ less disease spread (Boyd and Banzhaf, 2007)
  - Vultures in India: their collapse raised death rates >4%, more in cities (Frank and Sudarshan, 2024)
  - Birding: a newly recognized bird species is worth ≈$277 per visitor (Wardle, 2025)
- **Challenge:** Satellites can see green and blue space, but they cannot do population surveys
  - Counting animals takes ground data (*e.g.,* eBird, camera traps) (Wardle, 2025; Kohl et al., 2024)
  - Without data, hard to measure the function of animals in cities, much less their benefit



From Measuring Ecosystem Services to Building Natural Assets

Who has the ability, power, and money to make new urban natural assets?

Cities build most new natural assets, often with non-profit partners

- New natural assets come from restoration, conservation, or (for animals) reintroduction
  - Ex: Urban Resources Initiative in New Haven restores abandoned lots into parks and plants trees with communities (Chen et al., 2024)
- Usually the city, often with a non-profit or a business improvement district
  - Cities own the land (sidewalks, parks), but have limited budgets and coordination frictions with private actors (Toxopeus and Polzin, 2021)
  - Private park organizations (mostly non-profits) are >40% of park spending in St. Louis and Detroit (The Trust for Public Land, 2025)
- Insurers invest where nature lowers their claims
  - 16 Canadian insurers fund wetland restoration to reduce flood risk (Kousky and Burley, 2025)
  - Coral reef insurance in Quintana Roo, Mexico (Schelske et al., 2021)
- Barriers to more private money: coordinating public and private financiers, and knowing the private benefit (Toxopeus and Polzin, 2021)

Research can unlock financing by measuring who benefits

- **Public:** Benefit-cost analysis to win grants or justify municipal budgets
  - 2023 revisions to Circulars A-4 and A-94 addressed ecosystem services; rescinded, but still good guidance (Gillingham, 2025; Sarinsky and Schwartz, 2025)
  - Loss of bats lowers county property tax revenue $1.2M per year (3.8%) (Nakhmurina et al., 2026)
- **Private:** Identify who benefits, and by how much
  - Tree cover protects storefront revenue from heat. In the southern U.S., avoided losses cover the cost of 10% tree cover in ≈4 years (revenue is not profit) (Creel, 2026)
  - Heat lowers retail consumption (Lee and Zheng, 2025) and labor productivity (Dasgupta et al., 2024), trees lower cooling costs (Heris et al., 2021; Han et al., 2024)
- **Health is an untapped private incentive**
  - Health insurers hold a tremendous amount of money; Medicare and Medicaid are among the largest federal budget items
  - Insurers already invest in natural assets that lower risk (Kousky and Burley, 2025; Schelske et al., 2021)
  - But hospital visits, treatment, and mental health are the benefits we have measured least

Conclusion

- Large literature, but no shared definition or method, and benefits are rarely measured in dollars
- Natural asset $\rightarrow$ ecosystem function $\rightarrow$ benefit
- Natural assets' function in cooling, draining stormwater/flood control, providing a place to recreate and supporting health are fairly well established
  - What is missing: who benefits, and by how much
- Natural assets are unequally distributed across neighborhoods
- Urban animals are a gap, partly because we cannot count them from space
- Importance
  - Measuring public and private benefits can unlock money to build new natural assets $\rightarrow$ health may be the biggest untapped source

Questions?

andie.creel@yale.edu



References

D. Albouy, P. Christensen, and I. Sarmiento-Barbieri. Unlocking amenities: Estimating public good complementarity. *Journal of Public Economics*, 182:104110, Feb. 2020. ISSN 0047-2727. doi: 10.1016/j.jpubeco.2019.104110. URL https://www.sciencedirect.com/science/article/pii/S0047272719301720.

F. Baró, I. Palomo, G. Zulian, P. Vizcaino, D. Haase, and E. Gómez-Baggethun. Mapping ecosystem service capacity, flow and demand for landscape and urban planning: A case study in the Barcelona metropolitan region. *Land Use Policy*, 57:405–417, Nov. 2016. ISSN 0264-8377. doi: 10.1016/j.landusepol.2016.06.006. URL https://www.sciencedirect.com/science/article/pii/S0264837716305555.

J. Boyd and S. Banzhaf. What are ecosystem services? the need for standardized environmental accounting units. *Ecological Economics*, 63(2):616–626, 2007. doi: 10.1016/j.ecolecon.2007.01.002.

J. Bronnmann, V. Liebelt, F. Marder, J. Meya, and M. Quaas. The value of naturalness of urban green spaces: Evidence from a discrete choice experiment. *Land Economics*, 99(4):528–542, 2023. doi: 10.3368/le.99.4.062321-0072R1.

S. Chen, K. Lund, C. Murphy-Dunning, and K. C. Seto. More extremely hot days, more heat exposure and fewer cooling options for people of color in Connecticut, U.S. *npj Urban Sustainability*, 4(1):1–11, 2024. doi: 10.1038/s42949-024-00186-5. URL https://www.nature.com/articles/s42949-024-00186-5.

C. Conzelmann, A. Salazar Miranda, T. Phan, and J. Hoffman. Long-term effects of redlining on environmental risk exposure. *SSRN Scholarly Paper*, (4284307), 2022. doi: 10.21144/wp22-09.

A. M. Creel. The Value of Nature-Based Adaptation: Evidence that Urban Tree Cover Protects Revenue from Heat. *Working Paper*, 2026.

A. M. Creel and E. P. Fenichel. 20 Years of Local Outdoor Recreation and Its Value to Americans. *Working Paper*, 2026.

S. Dasgupta, E. J. Z. Robinson, S. Shayegh, F. Bosello, R. J. Park, and S. N. Gosling. Heat stress and the labour force. *Nature Reviews Earth & Environment*, 5 (12):859–872, 2024. ISSN 2662-138X. doi: 10.1038/s43017-024-00606-1. URL https://www.nature.com/articles/s43017-024-00606-1.

L. Diep and T. McPhearson. Empowering cities globally: Four levers for transformative urban adaptation with nature-based solutions. *Proceedings of the National Academy of Sciences*, 122(29):e2315912121, 2025. doi: 10.1073/pnas.2315912121. URL https://www.pnas.org/doi/10.1073/pnas.2315912121.

G. H. Donovan, D. T. Butry, Y. L. Michael, J. P. Prestemon, A. M. Liebhold, D. Gatziolis, and M. Y. Mao. The relationship between trees and human health: Evidence from the spread of the emerald ash borer. *American Journal of Preventive Medicine*, 44(2):139–145, 2013. doi: 10.1016/j.amepre.2012.09.066.

E. P. Fenichel. A new era of economic measurement for the environment and natural capital. *Review of Environmental Economics and Policy*, 18(2):321–330, 2024. doi: 10.1086/730513.

References

E. Frank and A. Sudarshan. The social costs of keystone species collapse: Evidence from the decline of vultures in India. *American Economic Review*, 114(10): 3007–3040, 2024. doi: 10.1257/aer.20230016.

E. G. Frank. The economic impacts of ecosystem disruptions: Costs from substituting biological pest control. *Science*, 385(6713):eadg0344, 2024. doi: 10.1126/science.adg0344. URL https://www.science.org/doi/abs/10.1126/science.adg0344.

K. T. Gillingham. Reforming regulatory and spending analysis: Introduction to a symposium on revising benefit–cost guidance. *Review of Environmental Economics and Policy*, 19(2):195–199, 2025. doi: 10.1086/735744. URL https://doi.org/10.1086/735744.

C. J. Gronlund, V. J. Berrocal, J. L. White-Newsome, K. C. Conlon, and M. S. O'Neill. Vulnerability to extreme heat by socio-demographic characteristics and area green space among the elderly in Michigan, 1990–2007. *Environmental Research*, 136:449–461, 2015. doi: 10.1016/j.envres.2014.08.042.

E. Gómez-Baggethun and D. N. Barton. Classifying and valuing ecosystem services for urban planning. *Ecological Economics*, 86:235–245, 2013. doi: 10.1016/j.ecolecon.2012.08.019.

R. Haines-Young and M. Potschin. The links between biodiversity, ecosystem services and human well-being. In D. G. Raffaelli and C. L. J. Frid, editors, *Ecosystem Ecology: A New Synthesis*, pages 110–139. Cambridge University Press, Cambridge, 2010.

Z. A. Hamstead, D. Fisher, R. T. Ilieva, S. A. Wood, T. McPhearson, and P. Kremer. Geolocated social media as a rapid indicator of park visitation and equitable park access. *Computers, Environment and Urban Systems*, 72:38–50, Nov. 2018. ISSN 01989715. doi: 10.1016/j.compenvurbsys.2018.01.007. URL https://linkinghub.elsevier.com/retrieve/pii/S0198971517303538.

L. Han, S. Heblich, C. Timmins, and Y. Zylberberg. Cool cities: The value of urban trees. *National Bureau of Economic Research*, 2024. doi: 10.3386/w32063. URL https://www.nber.org/papers/w32063.

M. Heris, K. J. Bagstad, C. Rhodes, A. Troy, A. Middel, K. G. Hopkins, and J. Matuszak. Piloting urban ecosystem accounting for the United States. *Ecosystem Services*, 48:101226, 2021. ISSN 2212-0416. doi: 10.1016/j.ecoser.2020.101226. URL https://www.sciencedirect.com/science/article/pii/S2212041620301686.

J. S. Hoffman, V. Shandas, and N. Pendleton. The effects of historical housing policies on resident exposure to intra-urban heat: A study of 108 US urban areas. *Climate*, 8(1):12, 2020. doi: 10.3390/cli8010012.

References

B. Janzen. Temperature and Mental Health: Evidence from Helpline Calls. *Journal of the Association of Environmental and Resource Economists*, 12(6): 1431–1457, 2025. ISSN 2333-5955. doi: 10.1086/736751. URL https://www.journals.uchicago.edu/doi/10.1086/736751.

G. D. Jenerette, S. L. Harlan, W. L. Stefanov, and C. A. Martin. Ecosystem services and urban heat riskscape moderation: water, green spaces, and social inequality in Phoenix, USA. *Ecological Applications*, 21(7):2637–2651, 2011. ISSN 1939-5582. doi: 10.1890/10-1493.1. URL https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1890/10-1493.1. _eprint: https://esajournals.onlinelibrary.wiley.com/doi/pdf/10.1890/10-1493.1.

V. Jennings and C. Gaither. Approaching environmental health disparities and green spaces: An ecosystem services perspective. *International Journal of Environmental Research and Public Health*, 12(2):1952–1968, 2015. doi: 10.3390/ijerph120201952.

B. M. Jesdale, R. Morello-Frosch, and L. Cushing. The Racial/Ethnic Distribution of Heat Risk–Related Land Cover in Relation to Residential Segregation. *Environmental Health Perspectives*, 121(7):811–817, July 2013. doi: 10.1289/ehp.1205919. URL https://ehp.niehs.nih.gov/doi/full/10.1289/ehp.1205919. Publisher: Environmental Health Perspectives.

A. Kaźmierczak. The contribution of local parks to neighbourhood social ties. *Landscape and Urban Planning*, 109(1):31–44, 2013. doi: 10.1016/j.landurbplan.2012.05.007.

B. L. Keeler, P. Hamel, T. McPhearson, M. H. Hamann, M. L. Donahue, K. A. Meza Prado, K. K. Arkema, G. N. Bratman, K. A. Brauman, J. C. Finlay, A. D. Guerry, S. E. Hobbie, J. A. Johnson, G. K. MacDonald, R. I. McDonald, N. Neverisky, and S. A. Wood. Social-ecological and technological factors moderate the value of urban nature. *Nature Sustainability*, 2(1):29–38, 2019. ISSN 2398-9629. doi: 10.1038/s41893-018-0202-1. URL https://www.nature.com/articles/s41893-018-0202-1.

H. A. Klaiber, J. K. Abbott, and V. K. Smith. Some Like It (Less) Hot: Extracting Trade-Off Measures for Physically Coupled Amenities. *Journal of the Association of Environmental and Resource Economists*, 4(4):1053–1079, 2017. ISSN 2333-5955. doi: 10.1086/692842. URL https://www.journals.uchicago.edu/doi/10.1086/692842.

M. T. Kohl, S. Fink, and T. Gelmi-Candusso. Quantifying urban wildlife habitat in a rapidly changing world. *PERC Research Plan*, 2024.

C. Kousky and T. Burley. Nature for insurance and insurance for nature. Technical report, Environmental Defense Fund, Aug. 2025.

S. Lee and S. Zheng. Extreme Temperatures, Adaptation Capacity, and Household Retail Consumption. *Journal of the Association of Environmental and Resource Economists*, 12(2):427–457, 2025. ISSN 2333-5955. doi: 10.1086/731056. URL https://www.journals.uchicago.edu/doi/10.1086/731056.

References

D. Locke, B. Hall, J. M. Grove, S. T. A. Pickett, L. A. Ogden, C. Aoki, C. G. Boone, and J. P. O'Neil-Dunne. Residential housing segregation and urban tree canopy in 37 US Cities. Technical report, SocArXiv, Jan. 2020. URL https://osf.io/preprints/socarxiv/97zcs/. type: article.

Y. Ma and J. Yang. A review of methods for quantifying urban ecosystem services. *Landscape and Urban Planning*, 253:105215, 2025. doi: 10.1016/j.landurbplan.2024.105215.

S. M. Maher, E. P. Fenichel, O. J. Schmitz, and W. L. Adamowicz. The economics of conservation debt: a natural capital approach to revealed valuation of ecological dynamics. *Ecological Applications*, 30(6):e02132, 2020. ISSN 1939-5582. doi: https://doi.org/10.1002/eap.2132. URL https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1002/eap.2132. _eprint: https://esajournals.onlinelibrary.wiley.com/doi/pdf/10.1002/eap.2132.

R. I. McDonald, T. Kroeger, P. Zhang, and P. Hamel. The value of US urban tree cover for reducing heat-related health impacts and electricity consumption. *Ecosystems*, 23(1):137–150, 2020. doi: 10.1007/s10021-019-00395-5.

A. Mohajerani, J. Bakaric, and T. Jeffrey-Bailey. The urban heat island effect, its causes, and mitigation, with reference to the thermal properties of asphalt concrete. *Journal of Environmental Management*, 197:522–538, 2017. doi: 10.1016/j.jenvman.2017.03.095.

A. Nakhmurina, D. T. Manning, and E. P. Fenichel. The fiscal impact of biodiversity loss and a pathway for conservation finance. *Science*, 391(6790):1110–1113, 2026. doi: 10.1126/science.ady0066.

F. Noack, D. Engist, and A. Larsen. The value of biodiversity: Evidence from migratory birds. *Working Paper*, 2025.

S. S. Ogletree, J.-H. Huang, D. Reif, L. Yang, C. Dunstan, N. Osakwe, J. I. Oh, and J. A. Hipp. The relationship between greenspace exposure and telomere length in the National Health and Nutrition Examination Survey. *Science of The Total Environment*, 905:167452, 2023. doi: 10.1016/j.scitotenv.2023.167452.

K. Peters, B. Elands, and A. Buijs. Social interactions in urban parks: Stimulating social cohesion? *Urban Forestry & Urban Greening*, 9(2):93–100, 2010. doi: 10.1016/j.ufug.2009.11.003.

J. L. Raynor, C. A. Grainger, and D. P. Parker. Wolves make roadways safer, generating large economic returns to predator conservation. *Proceedings of the National Academy of Sciences*, 118(22):e2023251118, 2021. doi: 10.1073/pnas.2023251118. URL https://www.pnas.org/doi/abs/10.1073/pnas.2023251118.

M. Sarinsky and J. A. Schwartz. The legal dynamics of rescinding the Circular A-4 update. Technical report, Institute for Policy Integrity, 2025.

References

O. Schelske, J. R. Bohn, and C. Fitzgerald. Insuring natural ecosystems as an innovative conservation funding mechanism: A case study on coral reefs. *Handbook of Disaster Risk Reduction for Resilience*, pages 435–452, 2021. doi: 10.1007/978-3-030-61278-8_19. URL https://link.springer.com/10.1007/978-3-030-61278-8_19.

N. Schwarz, M. Moretti, M. N. Bugalho, Z. G. Davies, D. Haase, J. Hack, A. Hof, Y. Melero, T. J. Pett, and S. Knapp. Understanding biodiversity-ecosystem service relationships in urban areas: A comprehensive literature review. *Ecosystem Services*, 27:161–171, 2017. doi: 10.1016/j.ecoser.2017.08.014. URL https://www.sciencedirect.com/science/article/pii/S2212041616305502.

K. R. E. Sims, L. G. Lee, N. Estrella-Luna, M. R. Lurie, and J. R. Thompson. Environmental justice criteria for new land protection can inform efforts to address disparities in access to nearby open space. *Environmental Research Letters*, 17(6):064014, 2022. ISSN 1748-9326. doi: 10.1088/1748-9326/ac6313. URL https://iopscience.iop.org/article/10.1088/1748-9326/ac6313.

P. Y. Tan, J. Zhang, M. Masoudi, J. B. Alemu, P. J. Edwards, A. Grêt-Regamey, D. R. Richards, J. Saunders, X. P. Song, and L. W. Wong. A conceptual framework to untangle the concept of urban ecosystem services. *Landscape and Urban Planning*, 200:103837, 2020. doi: 10.1016/j.landurbplan.2020.103837.

The Trust for Public Land. 2024 city park facts. https://www.tpl.org/city-park-facts, 2025. Blog; Accessed: 2025-08-28.

H. Toxopeus and F. Polzin. Reviewing financing barriers and strategies for urban nature-based solutions. *Journal of Environmental Management*, 289:112371, 2021. ISSN 0301-4797. doi: 10.1016/j.jenvman.2021.112371. URL https://www.sciencedirect.com/science/article/pii/S0301479721004333.

C. J. Veerkamp, A. M. Schipper, K. Hedlund, T. Lazarova, A. Nordin, and H. I. Hanson. A review of studies assessing ecosystem services provided by urban green and blue infrastructure. *Ecosystem Services*, 52:101367, 2021. doi: 10.1016/j.ecoser.2021.101367.

A. R. Wardle. The recreational value of rare species: Causal evidence from the Cassia crossbill. *Journal of the Association of Environmental and Resource Economists*, 12(5):1133–1164, 2025. doi: 10.1086/733759. URL https://www.journals.uchicago.edu/doi/10.1086/733759.

B. Wilson. Urban heat management and the legacy of redlining. *Journal of the American Planning Association*, 86(4):443–457, 2020. doi: 10.1080/01944363.2020.1759127.

N. H. Wong, C. L. Tan, D. D. Kolokotsa, and H. Takebayashi. Greenery as a mitigation and adaptation strategy to urban heat. *Nature Reviews Earth & Environment*, 2(3):166–181, 2021. ISSN 2662-138X. doi: 10.1038/s43017-020-00129-5. URL https://www.nature.com/articles/s43017-020-00129-5.